

Initiating Search

November 4, 2025, 4:47 PM

• Search:

CAS Lexicon Search:

Chan-Lam coupling reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (367)	 References	View Results
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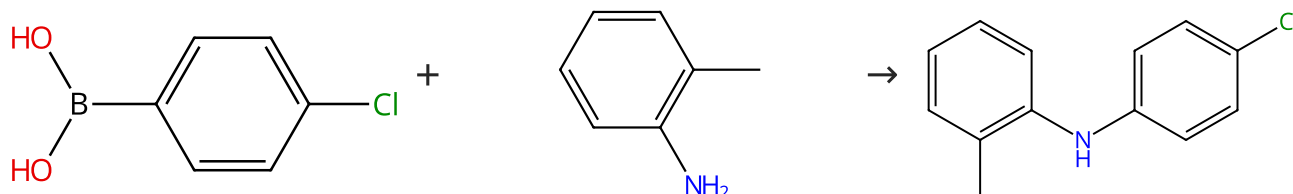
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (109) Suppliers (78) Suppliers (5)

31-146-CAS-18315227

Steps: 1 Yield: 100%

Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives

By: Yoo, Woo-Jin; et al

Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.

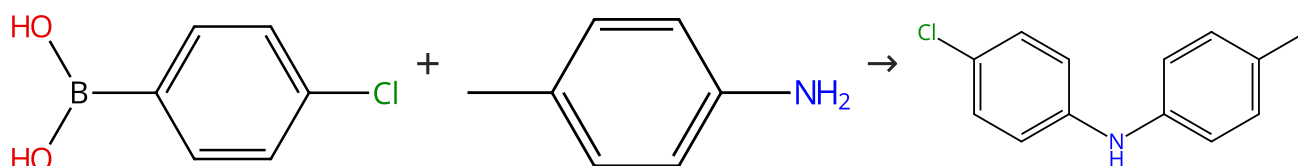
- 1.1 Reagents: 2,6-Lutidine
Catalysts: Cupric acetate, Myristic acid, *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C

- 1.2 Reagents: Silica

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (109) Suppliers (76) Suppliers (9)

31-146-CAS-18135981

Steps: 1 Yield: 100%

Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives

By: Yoo, Woo-Jin; et al

Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.

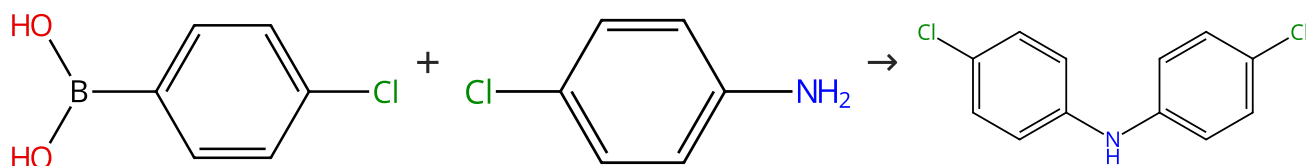
- 1.1 Reagents: 2,6-Lutidine
Catalysts: Cupric acetate, Myristic acid, *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C

- 1.2 Reagents: Silica

Experimental Protocols

Scheme 3 (1 Reaction)

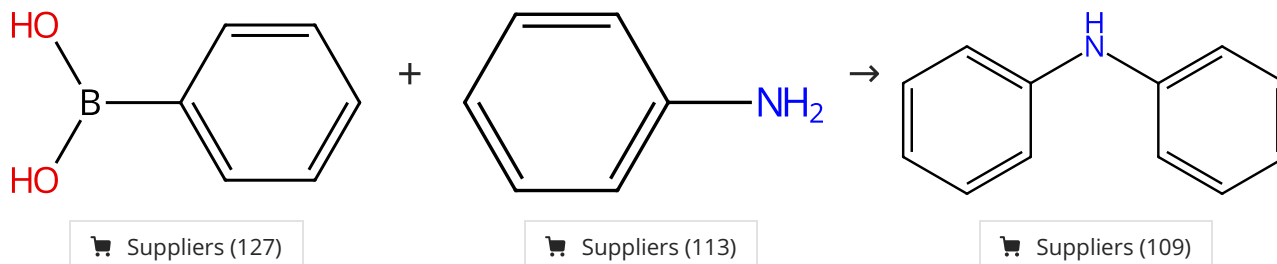
Steps: 1 Yield: 100%

 Suppliers (109) Suppliers (98) Suppliers (44)

31-146-CAS-18129460 Steps: 1 Yield: 100%	Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives
1.1 Reagents: 2,6-Lutidine Catalysts: Cupric acetate, Myristic acid, <i>fac</i> -Tris(2-(2-pyridinyl)phenyl)iridium Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C	By: Yoo, Woo-Jin; et al Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.
1.2 Reagents: Silica Experimental Protocols	

Scheme 4 (1 Reaction)

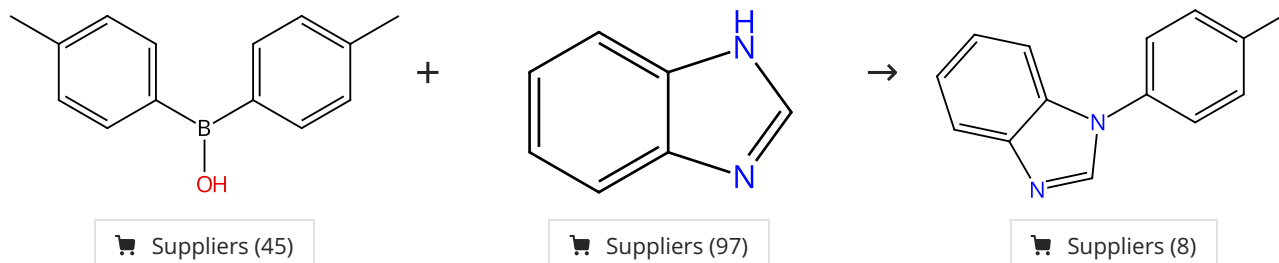
Steps: 1 Yield: 100%



31-146-CAS-18167067 Steps: 1 Yield: 100%	Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives
1.1 Reagents: 2,6-Lutidine Catalysts: Cupric acetate, Myristic acid, <i>fac</i> -Tris(2-(2-pyridinyl)phenyl)iridium Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C	By: Yoo, Woo-Jin; et al Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.
1.2 Reagents: Silica Experimental Protocols	

Scheme 5 (1 Reaction)

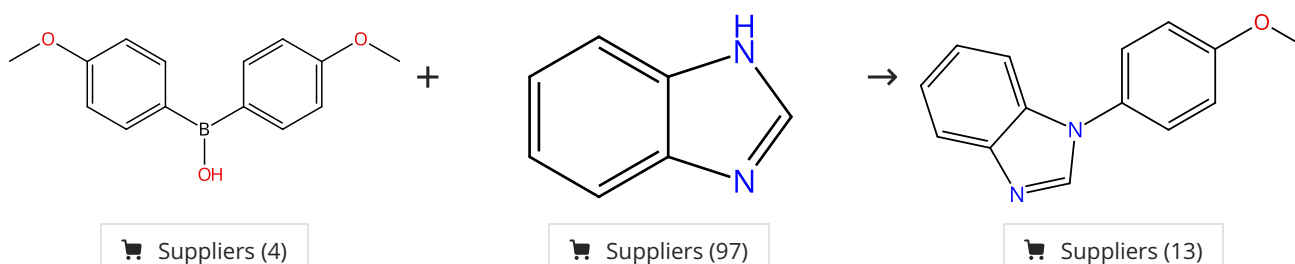
Steps: 1 Yield: 99%



31-146-CAS-21433081 Steps: 1 Yield: 99%	Base-assisted, copper-catalyzed N-arylation of (benz)imidazoles and amines with diarylboronic acids
1.1 Reagents: <i>N,N,N',N'</i> -Tetramethylethylenediamine Catalysts: Cupric acetate Solvents: Methanol; 4 h, rt	By: Guan, Changwei; et al Tetrahedron (2017), 73(49), 6906-6913.
Experimental Protocols	

Scheme 6 (1 Reaction)

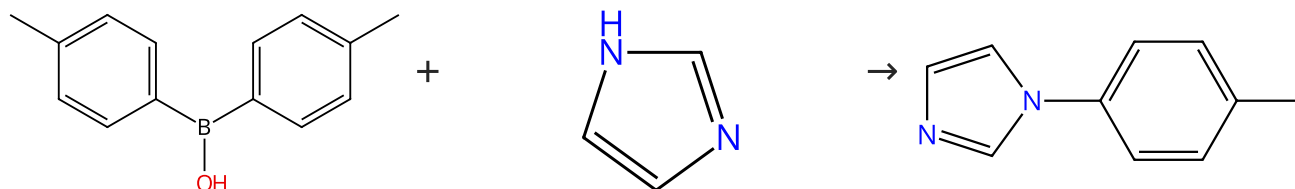
Steps: 1 Yield: 99%



31-146-CAS-21445022	Steps: 1 Yield: 99%	Base-assisted, copper-catalyzed N-arylation of (benz) imidazoles and amines with diarylborinic acids
1.1 Reagents: <i>N,N,N',N'</i> -Tetramethylethylenediamine Catalysts: Cupric acetate Solvents: Methanol; 4 h, rt		By: Guan, Changwei; et al Tetrahedron (2017), 73(49), 6906-6913.
Experimental Protocols		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (45)

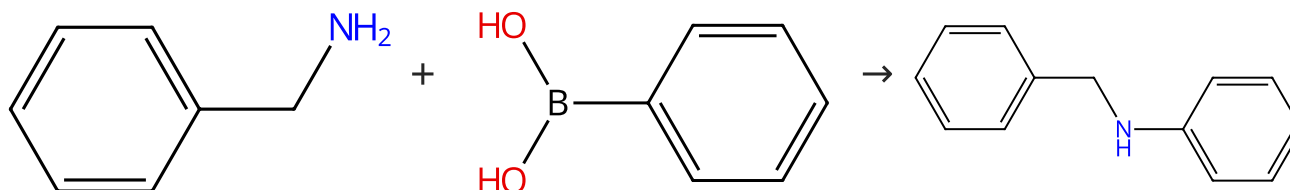
Suppliers (206)

Suppliers (23)

31-146-CAS-21373841	Steps: 1 Yield: 99%	Base-assisted, copper-catalyzed N-arylation of (benz) imidazoles and amines with diarylborinic acids
1.1 Reagents: <i>N,N,N',N'</i> -Tetramethylethylenediamine Catalysts: Cupric acetate Solvents: Methanol; 4 h, rt		By: Guan, Changwei; et al Tetrahedron (2017), 73(49), 6906-6913.
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (81)

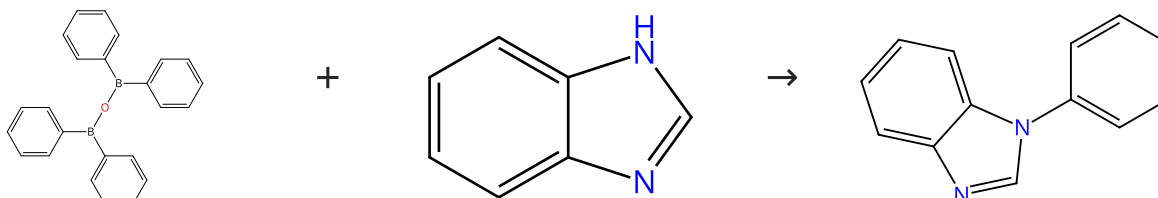
Suppliers (127)

Suppliers (90)

31-146-CAS-21445098	Steps: 1 Yield: 99%	Chan-Evans-Lam Couplings with Copper Iminoarylsulfonate Complexes: Scope and Mechanism
1.1 Reagents: Oxygen Catalysts: 2133472-58-7 Solvents: Methanol; 12 h, 50 °C		By: Hardouin Duparc, Valerie; et al ACS Catalysis (2018), 8(8), 7308-7325.
Experimental Protocols		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (44)

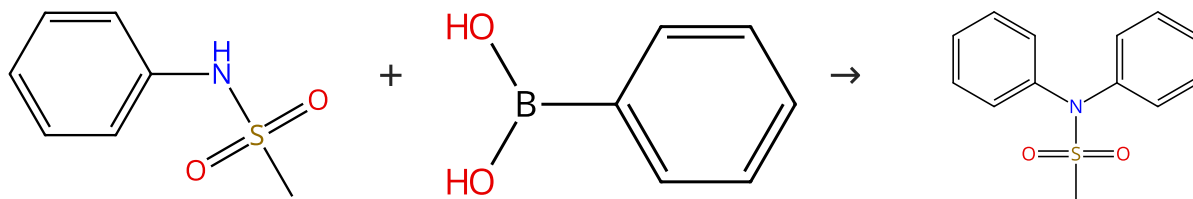
Suppliers (97)

Suppliers (49)

31-146-CAS-21434142	Steps: 1 Yield: 98%	Base-assisted, copper-catalyzed N-arylation of (benz) imidazoles and amines with diarylborinic acids
1.1 Reagents: <i>N,N,N',N'</i> -Tetramethylethylenediamine Catalysts: Cupric acetate Solvents: Methanol; 4 h, rt		By: Guan, Changwei; et al Tetrahedron (2017), 73(49), 6906-6913.
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (69)

Suppliers (127)

Suppliers (11)

31-146-CAS-21195289

Steps: 1 Yield: 98%

Mechanistic Insight Enables Practical, Scalable, Room Temperature Chan-Lam N-Arylation of N-Aryl Sulfonamides

By: Vantourout, Julien C.; et al

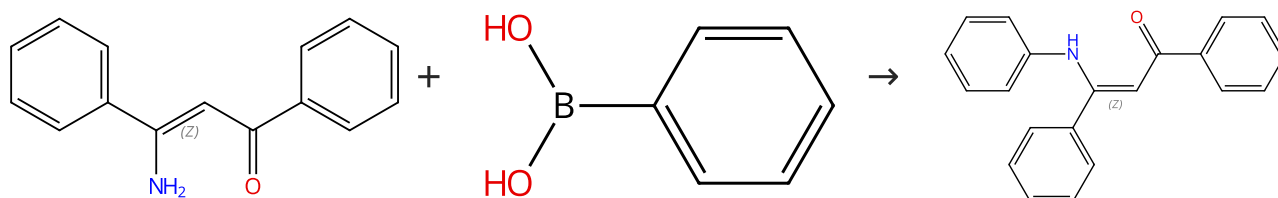
ACS Catalysis (2018), 8(10), 9560-9566.

1.1 **Reagents:** *N*-Methylpiperidine, Oxygen
Catalysts: Tetrakis(acetonitrile)copper(1+) hexafluoro phosphate
Solvents: Acetonitrile; 16 h, 20 - 25 °C

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Suppliers (9)

Suppliers (127)

Double bond geometry shown

Suppliers (4)

31-146-CAS-21170872

Steps: 1 Yield: 98%

Copper-promoted Chan-Lam coupling between enaminones and aryl boronic acids

By: Duan, Xiyan; et al

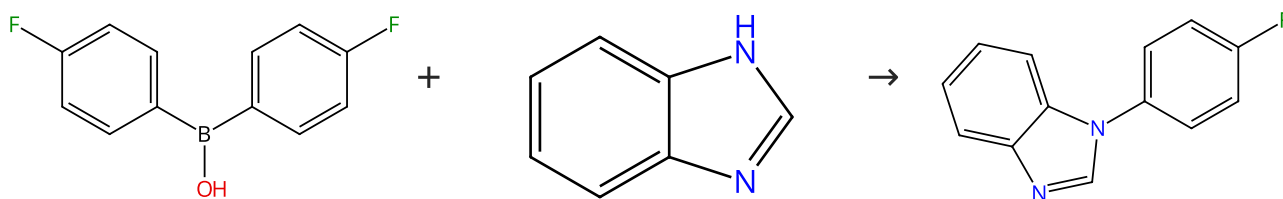
Tetrahedron Letters (2018), 59(47), 4187-4190.

1.1 **Catalysts:** Triethylamine, Cupric acetate
Solvents: 1,2-Dichloroethane; 24 h, 50 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (7)

Suppliers (97)

Suppliers (6)

31-146-CAS-21433865

Steps: 1 Yield: 98%

Base-assisted, copper-catalyzed N-arylation of (benz)imidazoles and amines with diarylboronic acids

By: Guan, Changwei; et al

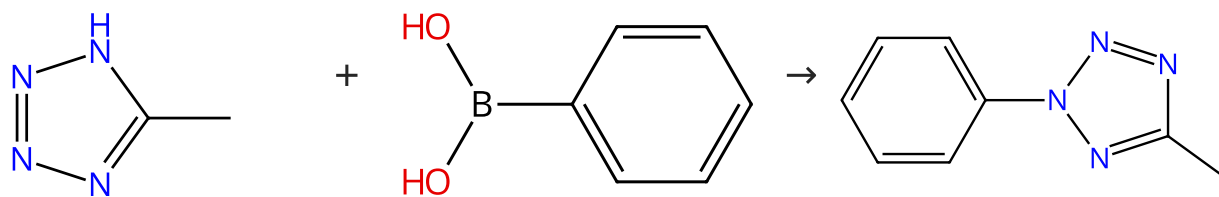
Tetrahedron (2017), 73(49), 6906-6913.

1.1 **Reagents:** *N,N,N,N*-Tetramethylethylenediamine
Catalysts: Cupric acetate
Solvents: Methanol; 4 h, rt

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (67)

Suppliers (127)

Suppliers (5)

31-146-CAS-21086875

Steps: 1 Yield: 98%

Chan-Evans-Lam Couplings with Copper Iminoarylsulfonate Complexes: Scope and Mechanism

By: Hardouin Duparc, Valerie; et al

ACS Catalysis (2018), 8(8), 7308-7325.

1.1 Reagents: Oxygen

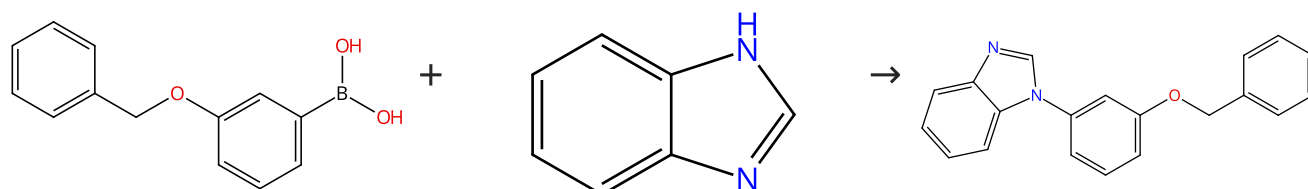
Catalysts: 2133472-58-7

Solvents: Methanol; 12 h, 50 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (97)

Supplier (1)

31-146-CAS-22161828

Steps: 1 Yield: 98%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

By: Jia, Xuefeng; et al

Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

1.1 Catalysts: 2252271-43-3

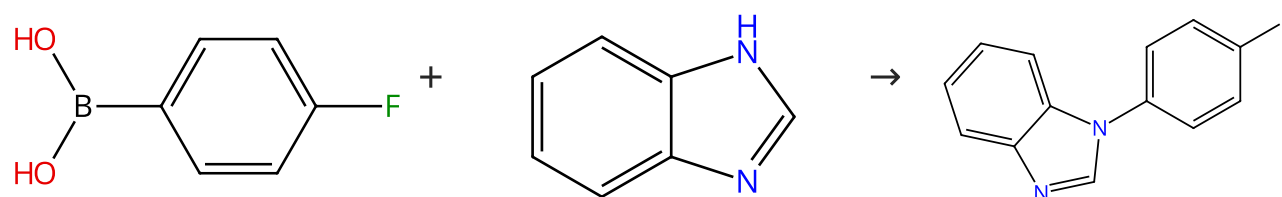
Solvents: Methanol; 22 h, rt

1.2 Reagents: Water; rt

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (101)

Suppliers (97)

Suppliers (6)

31-146-CAS-22161917

Steps: 1 Yield: 98%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

By: Jia, Xuefeng; et al

Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

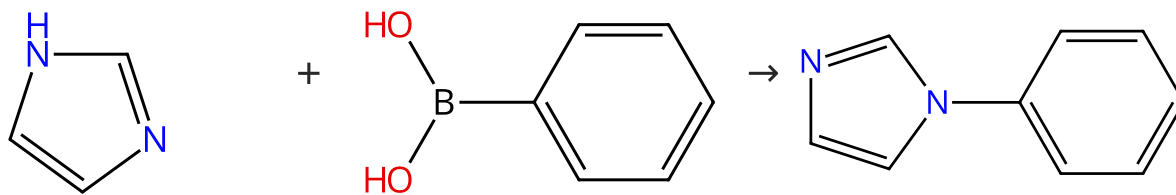
1.1 Catalysts: 2252271-43-3

Solvents: Methanol; 22 h, rt

1.2 Reagents: Water; rt

Scheme 16 (2 Reactions)

Steps: 1 Yield: 94-98%



Suppliers (206)

Suppliers (127)

Suppliers (80)

31-146-CAS-18058841

Steps: 1 Yield: 98%

A porous metal-organic framework as active catalyst for multiple C-N/C-C bond formation reactions

1.1 **Catalysts:** Bis[4-(1*H*-1,2,4-triazol-1-yl)benzoato-κ*O*,κ*O'*]copper
Solvents: Methanol; 24 h, 40 °C

By: Wang, Bin; et al

Experimental Protocols

Inorganic Chemistry Communications (2015), 61, 13-15.

31-146-CAS-1946787

Steps: 1 Yield: 94%

Unique copper-salen complex: an efficient catalyst for N-arylations of anilines and imidazoles at room temperature

1.1 **Reagents:** Potassium carbonate
Catalysts: Copper, [[2,2'-[(phenylmethylene)bis[(nitrilo-κ*N*)methylidyne]]bis[phenolato-κ*O*]](2-)]-
Solvents: Isopropanol; 10 h, 28 °C

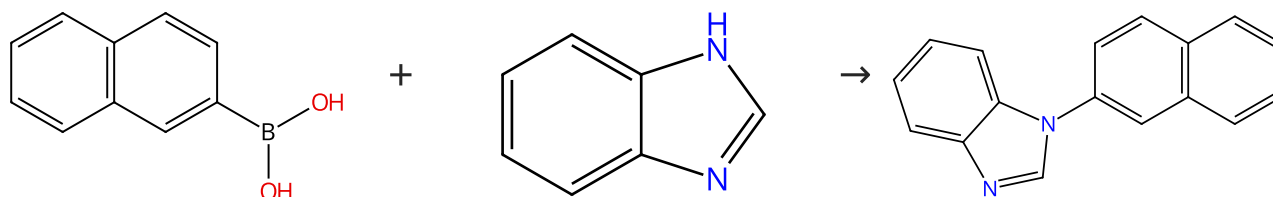
By: Gogoi, Ankur; et al

Tetrahedron Letters (2014), 55(1), 31-35.

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

Suppliers (97)

Suppliers (3)

31-146-CAS-21217150

Steps: 1 Yield: 97%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

1.1 **Reagents:** Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

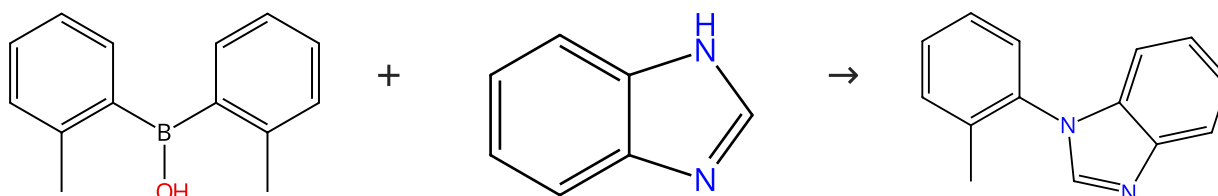
By: Musolino, Bryan J.; et al

Heterocycles (2015), 90(1), 271-297.

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (97)

Suppliers (3)

31-146-CAS-21434552

Steps: 1 Yield: 97%

Base-assisted, copper-catalyzed N-arylation of (benz)imidazoles and amines with diarylborinic acids

1.1 **Reagents:** *N,N,N',N'*-Tetramethylethylenediamine
Catalysts: Cupric acetate
Solvents: Methanol; 12 h, rt

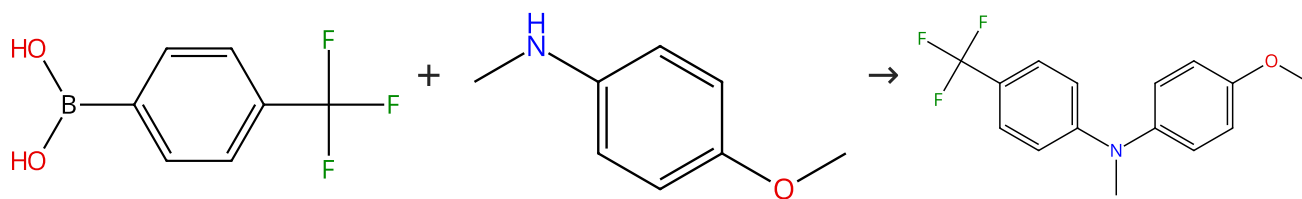
By: Guan, Changwei; et al

Tetrahedron (2017), 73(49), 6906-6913.

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (111)

Suppliers (82)

Suppliers (2)

31-146-CAS-21175554

Steps: 1 Yield: 97%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

By: Musolino, Bryan J.; et al

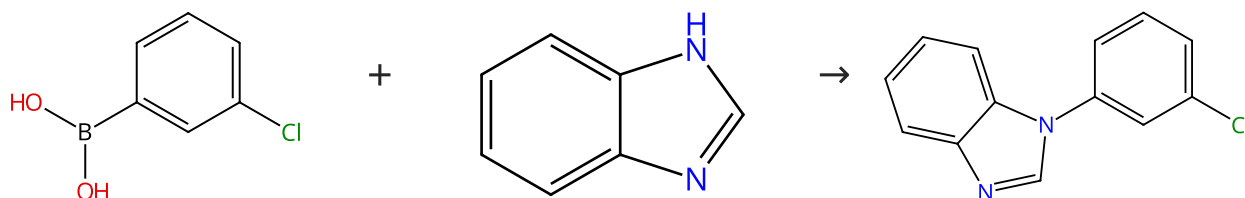
Heterocycles (2015), 90(1), 271-297.

1.1 Reagents: Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (111)

Suppliers (97)

Suppliers (3)

31-146-CAS-22162029

Steps: 1 Yield: 97%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

By: Jia, Xuefeng; et al

Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

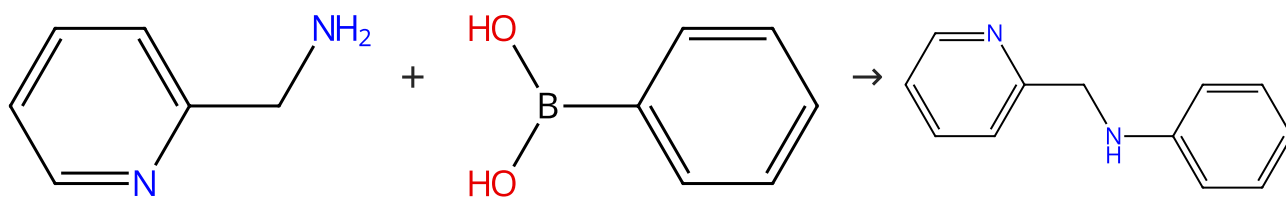
1.1 Catalysts: 2252271-43-3
Solvents: Methanol; 22 h, rt

1.2 Reagents: Water; rt

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (92)

Suppliers (127)

Suppliers (40)

31-146-CAS-21088748

Steps: 1 Yield: 97%

Chan-Evans-Lam Couplings with Copper Iminoaryl)sulfonate Complexes: Scope and Mechanism

By: Hardouin Duparc, Valerie; et al

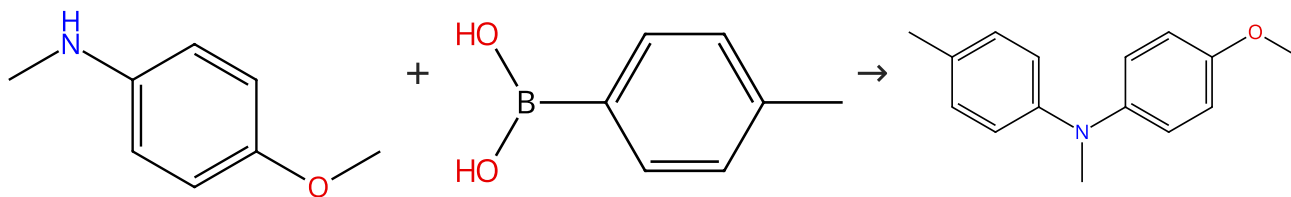
ACS Catalysis (2018), 8(8), 7308-7325.

1.1 Reagents: Oxygen
Catalysts: 2133472-58-7
Solvents: Methanol; 12 h, 50 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (82)

Suppliers (114)

Supplier (1)

31-146-CAS-21194490

Steps: 1 Yield: 96%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

1.1 Reagents: Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

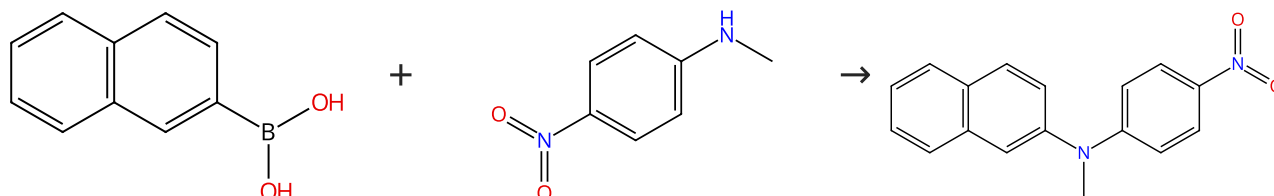
By: Musolino, Bryan J.; et al

Heterocycles (2015), 90(1), 271-297.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (102)

Suppliers (63)

31-146-CAS-21216684

Steps: 1 Yield: 96%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

1.1 Reagents: Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

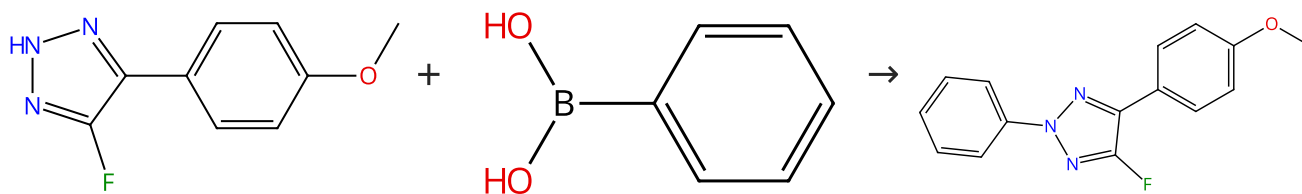
By: Musolino, Bryan J.; et al

Heterocycles (2015), 90(1), 271-297.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (127)

31-146-CAS-22105752

Steps: 1 Yield: 96%

Synthesis of 2,5-diaryl-4-halo-1,2,3-triazoles and comparative study of their fluorescent properties

1.1 Reagents: Oxygen
Catalysts: Cupric acetate
Solvents: Dimethyl sulfoxide; 1.5 - 2 h, 1 atm, rt → 100 °C

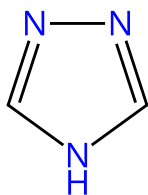
By: Motornov, Vladimir A.; et al

Tetrahedron (2018), 74(28), 3897-3903.

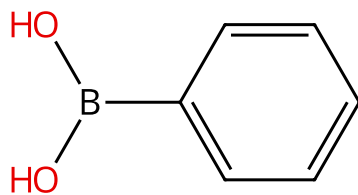
Experimental Protocols

Scheme 25 (1 Reaction)

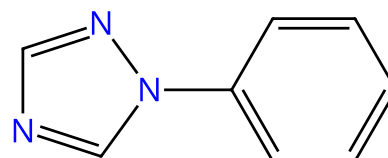
Steps: 1 Yield: 96%



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Suppliers (110)

Suppliers (127)

Suppliers (55)

31-146-CAS-22121526

Steps: 1 Yield: 96%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

1.1 Catalysts: 2252271-43-3
Solvents: Methanol; 22 h, rt

1.2 Reagents: Water; rt

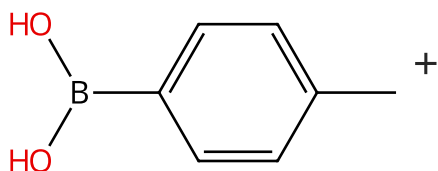
Experimental Protocols

By: Jia, Xuefeng; et al

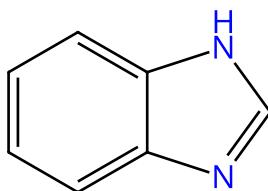
Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

Scheme 26 (1 Reaction)

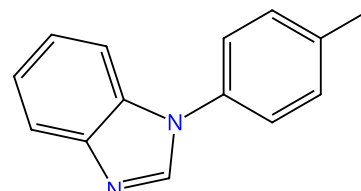
Steps: 1 Yield: 96%



+



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Suppliers (114)

Suppliers (97)

Suppliers (8)

31-146-CAS-22160584

Steps: 1 Yield: 96%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

1.1 Catalysts: 2252271-43-3
Solvents: Methanol; 22 h, rt

1.2 Reagents: Water; rt

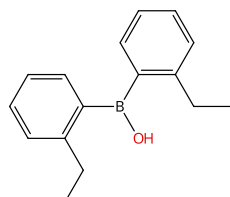
Experimental Protocols

By: Jia, Xuefeng; et al

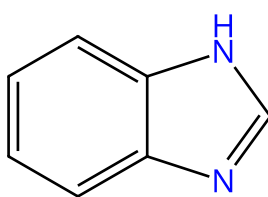
Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

Scheme 27 (1 Reaction)

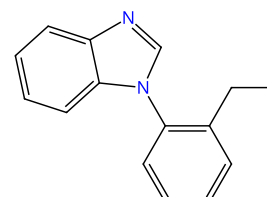
Steps: 1 Yield: 95%



+



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Supplier (1)

Suppliers (97)

31-146-CAS-22176556

Steps: 1 Yield: 95%

Base-assisted, copper-catalyzed N-arylation of (benz)imidazoles and amines with diarylboronic acids

1.1 Reagents: *N,N,N',N'*-Tetramethylethylenediamine
Catalysts: Cupric acetate
Solvents: Methanol; 12 h, rt

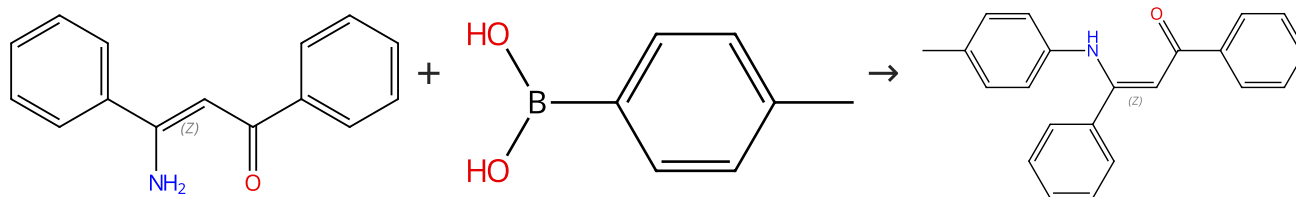
Experimental Protocols

By: Guan, Changwei; et al

Tetrahedron (2017), 73(49), 6906-6913.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 95%



Double bond geometry shown

Suppliers (114)

Double bond geometry shown

Supplier (1)

31-146-CAS-21110724

Steps: 1 Yield: 95%

Copper-promoted Chan-Lam coupling between enaminones and aryl boronic acids

1.1 **Catalysts:** Triethylamine, Cupric acetate
Solvents: 1,2-Dichloroethane; 24 h, 50 °C

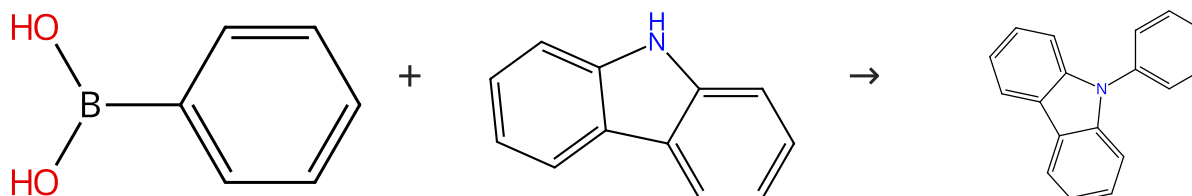
By: Duan, Xiyan; et al

Experimental Protocols

Tetrahedron Letters (2018), 59(47), 4187-4190.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (127)

Suppliers (117)

Suppliers (84)

31-146-CAS-21142829

Steps: 1 Yield: 95%

Cu(II) complex of pyridine-based polydentate as a novel, efficient, and highly reusable catalyst in C-N bond-forming reaction

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Copper, [*N,N*-bis[2-(hydroxy- κ O)phenyl]-2,6-pyridinedicarboxamidato(2-)- κ N¹, κ O², κ O⁶]-; 2.5 h, 120 °C

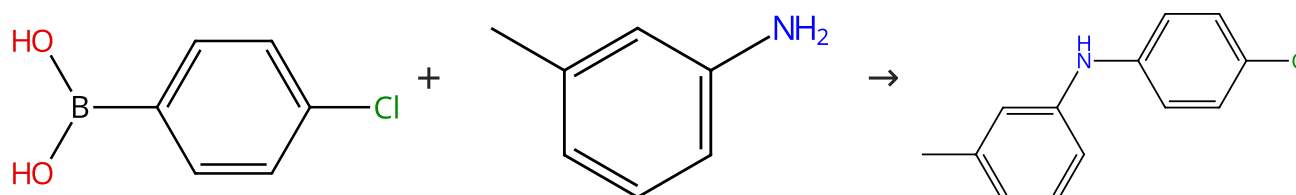
By: Sharghi, Hashem; et al

Experimental Protocols

Molecular Diversity (2017), 21(4), 855-864.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (109)

Suppliers (73)

Supplier (1)

31-146-CAS-18138188

Steps: 1 Yield: 95%

Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives

1.1 **Reagents:** 2,6-Lutidine
Catalysts: Cupric acetate, Myristic acid, *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C

By: Yoo, Woo-Jin; et al

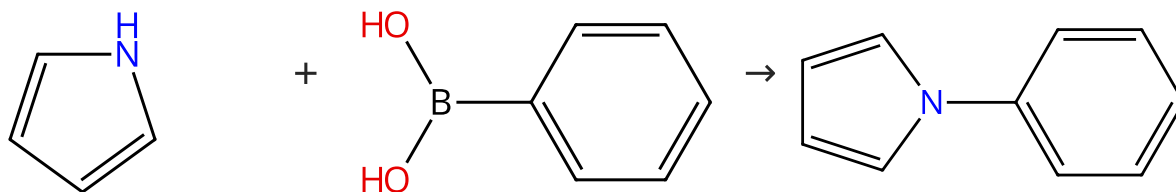
1.2 **Reagents:** Silica

Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (76)

Suppliers (127)

Suppliers (71)

31-146-CAS-21142111

Steps: 1 Yield: 95%

Cu(II) complex of pyridine-based polydentate as a novel, efficient, and highly reusable catalyst in C-N bond-forming reaction

By: Sharghi, Hashem; et al

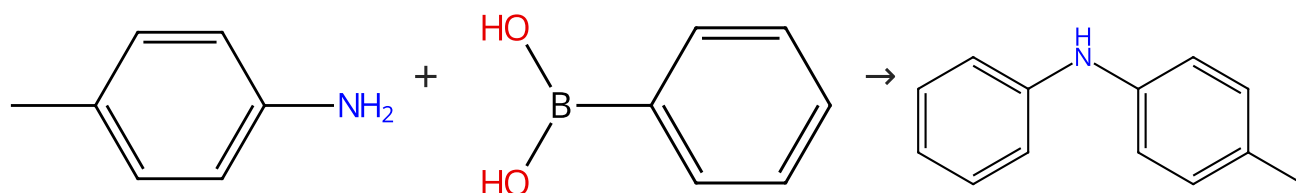
Molecular Diversity (2017), 21(4), 855-864.

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Copper, [*N,N'*-bis[2-(hydroxy- κO)phenyl]-2,6-pyridinedicarboxamidato(2-)- $\kappa N^1, \kappa O^2, \kappa O^6$]-; 1.5 h, 120 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (76)

Suppliers (127)

Suppliers (65)

31-146-CAS-11179456

Steps: 1 Yield: 95%

Unique copper-salen complex: an efficient catalyst for N-arylations of anilines and imidazoles at room temperature

By: Gogoi, Ankur; et al

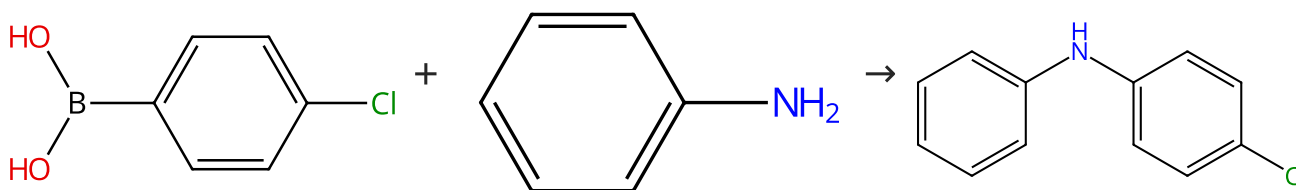
Tetrahedron Letters (2014), 55(1), 31-35.

1.1 **Reagents:** Potassium carbonate
Catalysts: Copper, [[2,2'-[(phenylmethylene)bis[(nitrilo- κN)methylidyne]]bis[phenolato- κO]](2-)]-
Solvents: Water; 12 h, 28 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (109)

Suppliers (113)

Suppliers (49)

31-146-CAS-18133823

Steps: 1 Yield: 95%

Visible-Light-Mediated Chan-Lam Coupling Reactions of Aryl Boronic Acids and Aniline Derivatives

By: Yoo, Woo-Jin; et al

Angewandte Chemie, International Edition (2015), 54(22), 6587-6590.

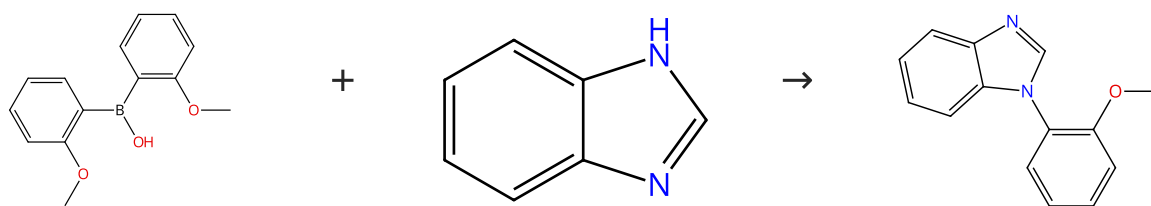
1.1 **Reagents:** 2,6-Lutidine
Catalysts: Cupric acetate, Myristic acid, *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: Acetonitrile, Toluene; 20 h, rt → 35 °C

1.2 **Reagents:** Silica

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (3)

Suppliers (97)

Suppliers (10)

31-146-CAS-21435629

Steps: 1 Yield: 94%

Base-assisted, copper-catalyzed N-arylation of (benz) imidazoles and amines with diarylborinic acids
1.1 Reagents: *N,N,N',N'*-Tetramethylethylenediamine

Catalysts: Cupric acetate

Solvents: Methanol; 12 h, rt

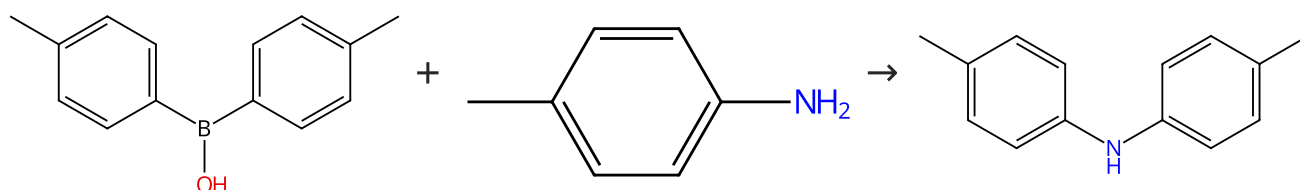
By: Guan, Changwei; et al

Tetrahedron (2017), 73(49), 6906-6913.

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (45)

Suppliers (76)

Suppliers (82)

31-146-CAS-21372731

Steps: 1 Yield: 94%

Base-assisted, copper-catalyzed N-arylation of (benz) imidazoles and amines with diarylborinic acids

1.1 Reagents: Pyridine

Catalysts: Cupric acetate

Solvents: Dichloromethane; 24 h, rt

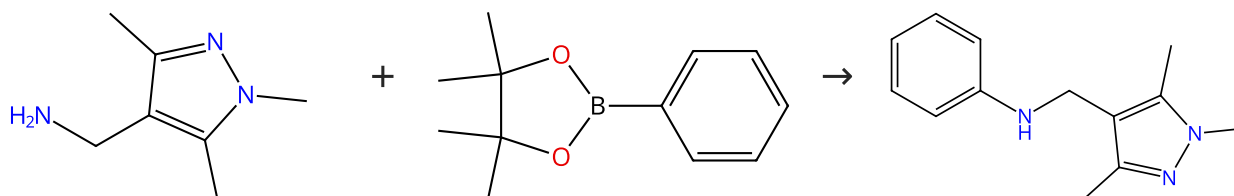
By: Guan, Changwei; et al

Tetrahedron (2017), 73(49), 6906-6913.

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (62)

Suppliers (80)

Suppliers (21)

31-146-CAS-21261580

Steps: 1 Yield: 94%

Spectroscopic Studies of the Chan-Lam Amination: A Mechanism-Inspired Solution to Boronic Ester Reactivity
1.1 Reagents: Cupric acetate, Oxygen, Boric acid (H_3BO_3)

Solvents: Acetonitrile; 24 h, 80 °C

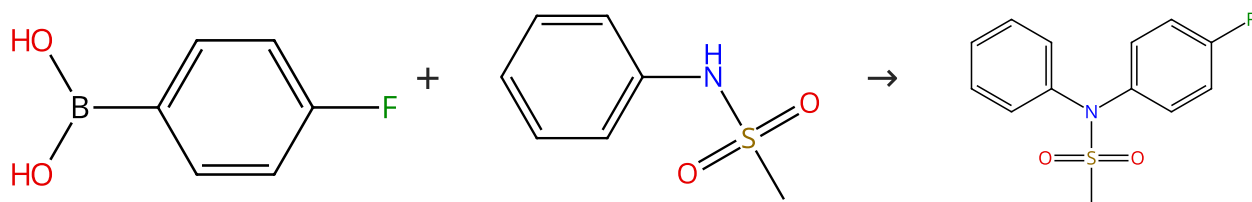
By: Vantourout, Julien C.; et al

Journal of the American Chemical Society (2017), 139(13), 4769-4779.

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (101)

Suppliers (69)

Supplier (1)

31-146-CAS-22117502

Steps: 1 Yield: 94%

Mechanistic Insight Enables Practical, Scalable, Room Temperature Chan-Lam N-Arylation of N-Aryl Sulfonamides

1.1 **Reagents:** *N*-Methylpiperidine, Oxygen
Catalysts: Tetrakis(acetonitrile)copper(1+) hexafluoro phosphate
Solvents: Acetonitrile; 72 h, 20 - 25 °C

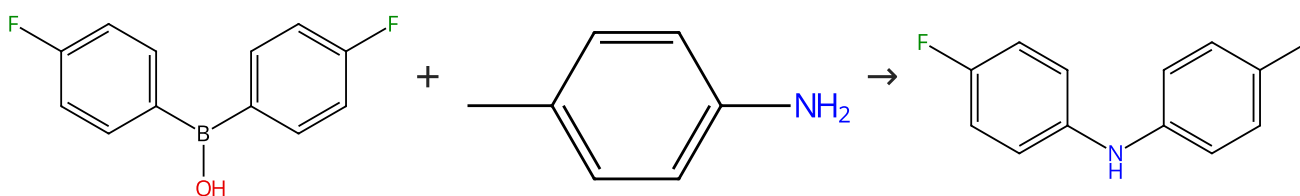
By: Vantourout, Julien C.; et al

ACS Catalysis (2018), 8(10), 9560-9566.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (7)

Suppliers (76)

Suppliers (5)

31-146-CAS-21385874

Steps: 1 Yield: 94%

Base-assisted, copper-catalyzed N-arylation of (benz)imidazoles and amines with diarylboronic acids

1.1 **Reagents:** Pyridine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 24 h, rt

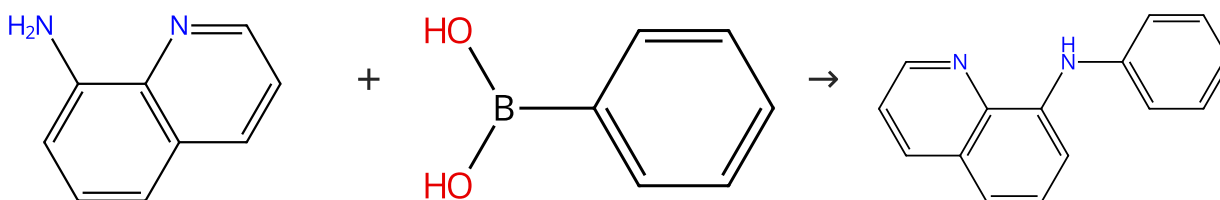
By: Guan, Changwei; et al

Tetrahedron (2017), 73(49), 6906-6913.

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (91)

Suppliers (127)

Suppliers (6)

31-146-CAS-18095265

Steps: 1 Yield: 94%

Triazine-based mesoporous covalent imine polymers as solid supports for copper-mediated Chan-Lam cross-coupling N-arylation reactions

1.1 **Reagents:** Triethylamine, Oxygen
Catalysts: Copper, Benzaldehyde, 4,4',4''-[1,3,5-triazine-2,4,6-triyltris(oxy)]tris-, polymer with 1,4-benzenediamine
Solvents: 1,2-Dichloroethane; 16 h, rt

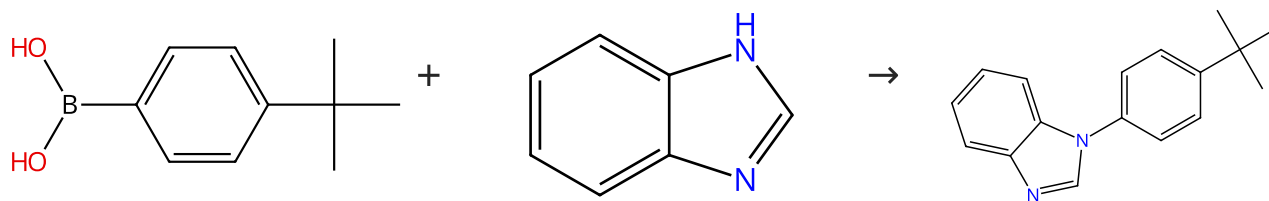
By: Puthiaraj, Pillaiyar; et al

Chemistry - A European Journal (2014), 20(28), 8761-8770.

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (92)

Suppliers (97)

Suppliers (3)

31-146-CAS-22160476

Steps: 1 Yield: 94%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

By: Jia, Xuefeng; et al

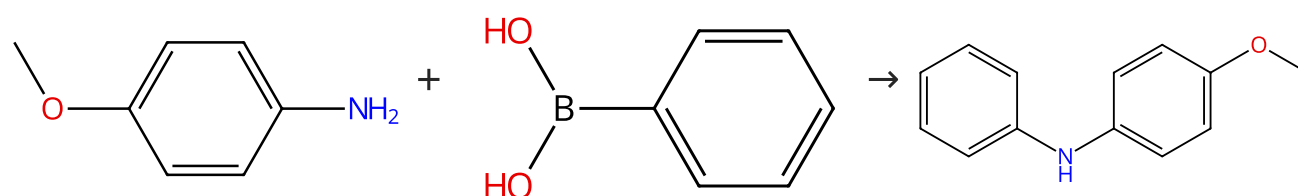
Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

- 1.1 Catalysts: 2252271-43-3
Solvents: Methanol; 22 h, rt
- 1.2 Reagents: Water; rt

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (83)

Suppliers (127)

Suppliers (69)

31-146-CAS-10898203

Steps: 1 Yield: 94%

Unique copper-salen complex: an efficient catalyst for N-arylations of anilines and imidazoles at room temperature

By: Gogoi, Ankur; et al

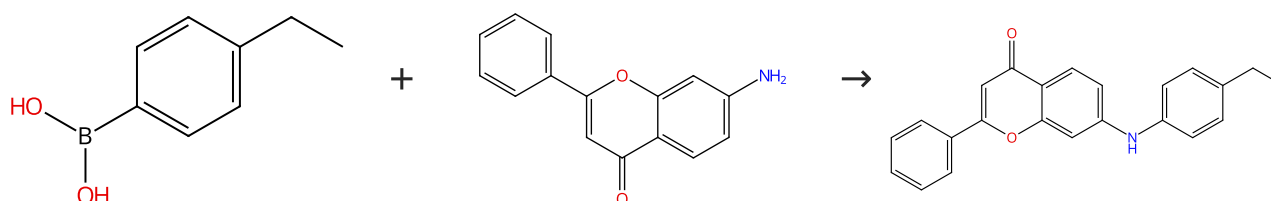
Tetrahedron Letters (2014), 55(1), 31-35.

- 1.1 Reagents: Potassium carbonate
Catalysts: Copper, [[2,2'-[(phenylmethylene)bis[(nitrilo-κN)methylidene]]bis[phenolato-κO]](2-)]-
Solvents: Water; 16 h, 28 °C

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (87)

Suppliers (33)

31-146-CAS-18037922

Steps: 1 Yield: 93%

Triazine-based mesoporous covalent imine polymers as solid supports for copper-mediated Chan-Lam cross-coupling N-arylation reactions

By: Puthiaraj, Pillaiyar; et al

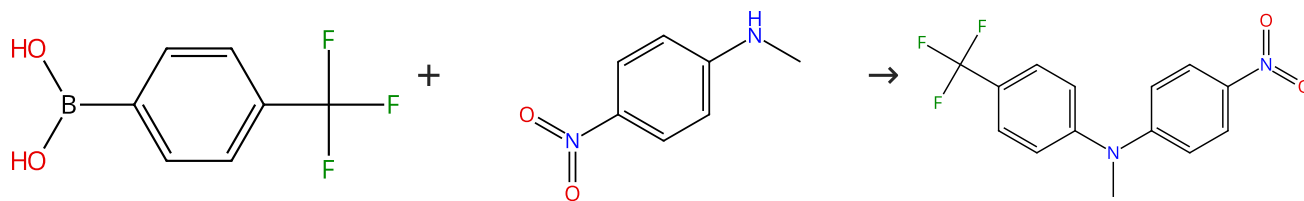
Chemistry - A European Journal (2014), 20(28), 8761-8770.

- 1.1 Reagents: Triethylamine, Oxygen
Catalysts: Copper, Benzaldehyde, 4,4',4''-[1,3,5-triazine-2,4,6-triyltris(oxy)]tris-, polymer with 1,4-benzenediamine
Solvents: 1,2-Dichloroethane; 16 h, rt

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (111)

Suppliers (63)

31-146-CAS-21228719

Steps: 1 Yield: 93%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

By: Musolino, Bryan J.; et al

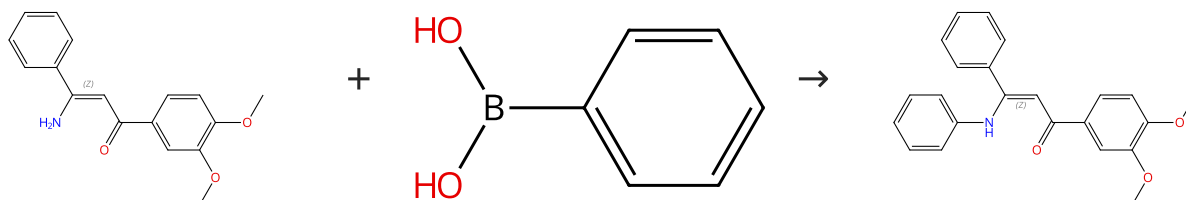
Heterocycles (2015), 90(1), 271-297.

1.1 **Reagents:** Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 93%



Double bond geometry shown

Suppliers (127)

Double bond geometry shown

31-146-CAS-22093958

Steps: 1 Yield: 93%

Copper-promoted Chan-Lam coupling between enaminones and aryl boronic acids

By: Duan, Xiyan; et al

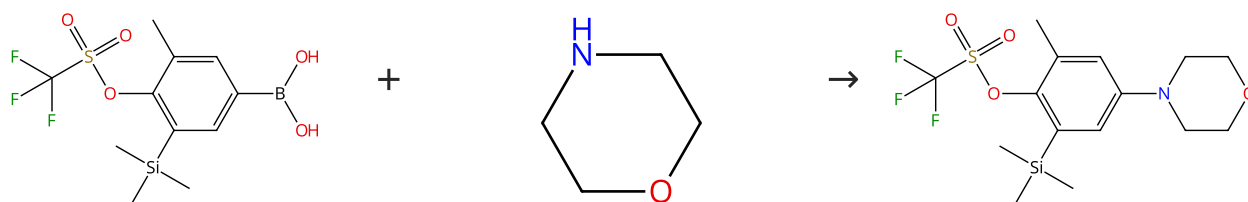
Tetrahedron Letters (2018), 59(47), 4187-4190.

1.1 **Catalysts:** Triethylamine, Cupric acetate
Solvents: 1,2-Dichloroethane; 24 h, 50 °C

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (88)

31-146-CAS-18314444

Steps: 1 Yield: 93%

Boryl (Hetero)aryne Precursors as Versatile Arylation Reagents: Synthesis through C-H Activation and Orthogonal Reactivity

By: Demory, Emilien; et al

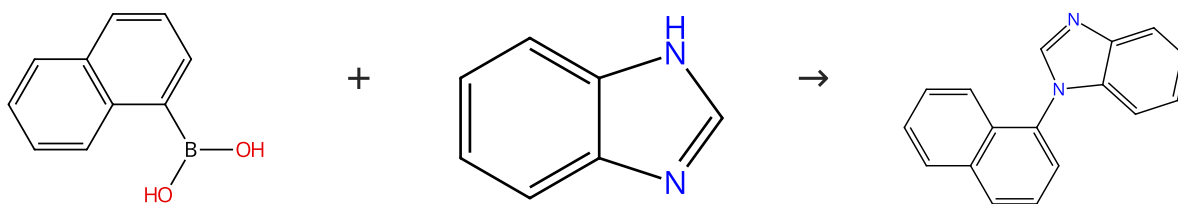
Angewandte Chemie, International Edition (2015), 54(40), 11765-11769.

1.1 **Reagents:** Oxygen
Catalysts: Cupric acetate
Solvents: Dichloromethane; 5 min, 20 °C
 1.2 20 h, 40 °C

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (99)

Suppliers (97)

Suppliers (3)

31-146-CAS-22122464

Steps: 1 Yield: 93%

N,O-Bidentate ligand-tunable copper(II) complexes as a catalyst for Chan-Lam coupling reactions of arylboronic acids with 1H-imidazole derivatives

By: Jia, Xuefeng; et al

Organic & Biomolecular Chemistry (2018), 16(46), 8984-8988.

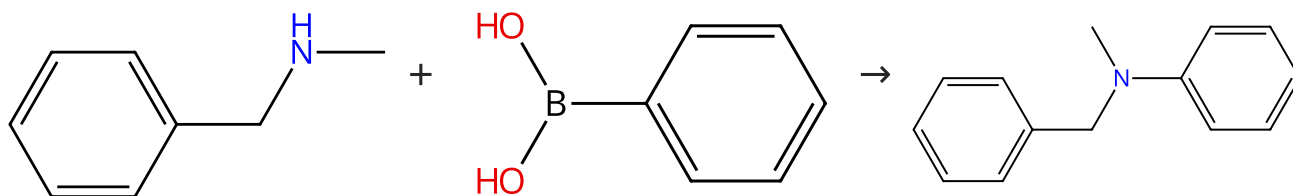
1.1 **Catalysts:** 2252271-43-3
Solvents: Methanol; 22 h, rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (84)

Suppliers (127)

Suppliers (55)

31-146-CAS-21068187

Steps: 1 Yield: 93%

A quick Chan-Lam C-N and C-S cross coupling at room temperature in the presence of square pyramidal [Cu(DMAP)₄]I as a catalyst

By: Roy, Subhasish; et al

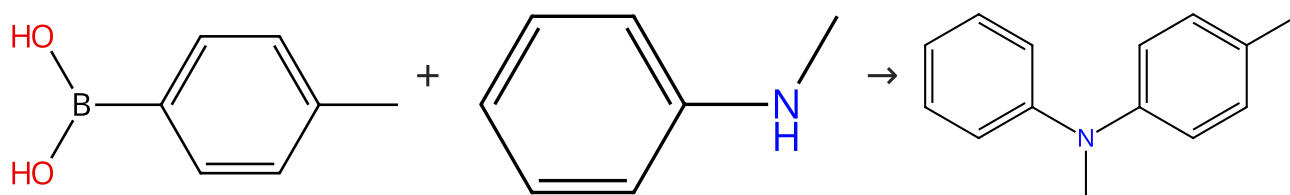
Chemical Communications (Cambridge, United Kingdom) (2016), 52(6), 1170-1173.

1.1 **Catalysts:** Copper(1+), tetrakis(*N,N*-dimethyl-4-pyridinamine-κ^N)iodo-, iodide (1:1), (*SP*-5-12)-
Solvents: Methanol; 5 min, rt

Experimental Protocols

Scheme 48 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (114)

Suppliers (73)

Suppliers (25)

31-146-CAS-21193013

Steps: 1 Yield: 93%

Copper mediated formation of carbon-heteroatom bonds using organoboron reagents and ultrasound

By: Musolino, Bryan J.; et al

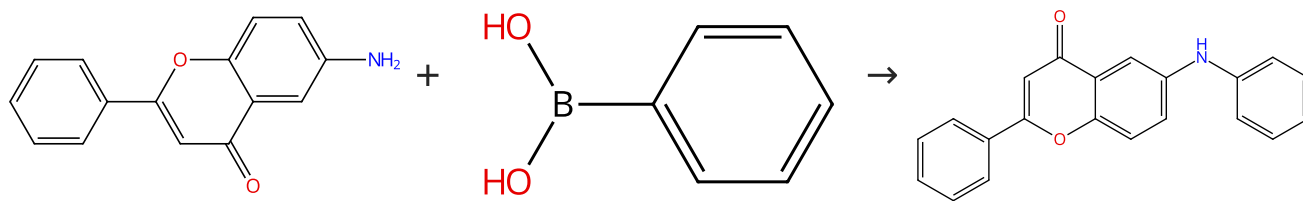
Heterocycles (2015), 90(1), 271-297.

1.1 **Reagents:** Triethylamine
Catalysts: Cupric acetate
Solvents: Dichloromethane; 4 h, 0 - 5 °C

Experimental Protocols

Scheme 49 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (34)

Suppliers (127)

Supplier (1)

31-146-CAS-51396

Steps: 1 Yield: 93%

1.1 **Reagents:** Triethylamine, Oxygen
Catalysts: Copper, Benzaldehyde, 4,4',4''-[1,3,5-triazine-2,4,6-triyltris(oxy)]tris-, polymer with 1,4-benzenediamine
Solvents: 1,2-Dichloroethane; 16 h, rt

Experimental Protocols

Triazine-based mesoporous covalent imine polymers as solid supports for copper-mediated Chan-Lam cross-coupling N-arylation reactions

By: Puthiaraj, Pillaiyar; et al

Chemistry - A European Journal (2014), 20(28), 8761-8770.